

surrounding the spinal cord. A local anesthetic is injected into the needle and flows through the tube into the epidural space so that the nerve roots carrying the pain stimulus to the brain are coated with anesthetic and pain is reduced or completely blocked.

This will affect sensation in your legs so that you need to remain in bed and your baby will be closely monitored. A stronger epidural also affects sensation in your bladder, so you will need to have a catheter. Used late in labor, a stronger epidural may mean that you need help to push the baby out, since the pelvic floor muscles will be heavy and ineffective. In this case, your doctor will put a hand on your abdomen to feel when a contraction starts and will tell you when to push. In some cases, an assisted delivery (see [Assisted birth](#)) becomes necessary.

When are epidurals used? Epidurals can be used throughout labor, but they're typically given during active labor when a woman is 4–5 cm dilated. Since everyone has varying pain thresholds, the time when one is requested varies. There are factors to bear in mind should you opt for an epidural late in labor. To minimize the risks, you must remain completely still during the placement of the epidural tube. If your labor has progressed too far to enable you to do this, the anesthesiologist may be unable to proceed with an epidural for your own interest. Also, if you choose to have an epidural late in labor, it may be necessary to give a high dose so that it takes effect in time, which has disadvantages (see [The pros and cons of an Epidural](#)).

If you're considering using an epidural, inform your doctor early in labor so that she can consult the anesthesiologist. The anesthesiologist may then discuss this with you and take a brief medical history to ensure that it's safe for you to have an epidural. She will discuss any risks, and answer any questions that you or your partner has, all of which can save time later on if you decide to go ahead.

There are occasions when an epidural is not advised. These include cases where a woman has had spinal surgery or is taking blood-thinning medication. Rarely, a woman may have an infection that could be exacerbated by an epidural.

Side effects There are a number of minor side effects. The medication can cause [blood pressure](#) to fall, so this will be monitored. If it does fall, you'll be given fluids and medication, and subsequent doses may be reduced.

It's common to experience itching with epidurals, caused by the release of histamine from the opioid component of a mobile epidural. Histamine is a substance released by the body during an allergic reaction that can cause itching. The itch can be treated, but in most cases it gets better on its own. If you develop an itch, a greater concentration of local anesthetic alone will be used.

It's not unusual to shiver with an epidural, although this is a more common side effect if a concentrated local anesthetic is used, as is the case for a cesarean delivery.

Epidural pain relief can cause a rise in temperature. If this occurs, you'll have a blood test to eliminate an infection since this can also cause your temperature to rise. You will be given preventative antibiotics while waiting for the blood test results, and acetaminophen to bring your temperature back to normal.

Problems with epidurals In addition to side effects, there can occasionally be problems with the effectiveness of an epidural. The anesthetic may not spread evenly in the epidural space, which may be caused by the epidural tube sitting on one side of the epidural space. This can mean that pain relief only occurs on one side of the body. If this occurs, the anesthesiologist will try to reposition the tube and give another dose of anesthetic. If this doesn't work, the only other solution is to redo the entire epidural.

Sometimes, one spot can remain painful, usually in the groin area or low down in the front of the abdomen, which is referred to as a "missed segment." This results from a single nerve root not being coated with the local anesthetic. Again, the anesthesiologist may reposition the tube. Sometimes, a stronger local anesthetic or an opioid is used to